25901111 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBTG202 - TISSUE CULTURE TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any **10** questions

Each question carries **2** marks

1. Which equipment is used in a plant tissue culture laboratory for ensuring sterile conditions during inoculation? [K] / [CO1]
2. What are the common challenges faced during meristem culture? [K] / [CO3]
3. Write any two steps in vaccine production. [K] / [CO4]
4. What is the role of Auxins in PTC? [U] / [CO2]
5. What are feeder layers? [K] / [CO3]
6. List any two commercial applications of tissue culture techniques. [K] / [CO3]
7. Name the process of formation of organs from callus. [K] / [CO3]
8. Which serum protein helps in cell attachment in animal tissue culture? [U] / [CO2]

9. Define "Sub-culturing.". [K] / [CO4]

10. Which equipment is used to observe cultured cells in animal cell culture lab? [K] / [CO1]

11. Why is a CO2 incubator used for some cultures? [U] / [CO1]

12. Define haploid production in plants. [K] / [CO3]

[2x10 = 20]

Part B

Short Essay

Answer any **5** questions

Each question carries **4** marks

13. Explain the concept of anchorage dependence in cultured cells.. [U] / [CO4]

14. What are the limitations of somaclonal variations? [K] / [CO3]

15. Comment on the importance of an incubation room in a plant tissue culture laboratory. [U] / [CO1]

16. Why is moist heat more effective than dry heat? [K] / [CO2]

17. Explain protoplast fusion. [K] / [CO3]

18. Describe how proper sterilization contributes to reproducible animal cell culture experiments. [U] / [CO2]

19. Write short essay on cytodifferentiation. [K] / [CO3]

[4x5 = 20]

Part C

Essay type questions

Answer any **1** questions

Each question carries **10** marks

20. Discuss the role of organogenesis in plant micropropagation and its benefits for agriculture. [U] / [CO3]

21. Comment on the historical development of plant biotechnology from selective breeding to genetic engineering. [U] / [CO1]

[10x1 = 10]